

PRISM-4D GLP-1R / Aleniglipron M2 Executive Dossier

Boardroom-facing overview of the M2 release

Confidential Discussion Material

Campaign	glp1r_aleniglipron
Release	PRISM_GLP1R_M2_DELIVERABLES_v1.1
Generated (UTC)	2026-05-24T03:47:26Z
Report type	Executive strategy and program-leadership review
CBOM Merkle root	1461e688bada66c7537bbce30e3b82ab1ec5f214e85cfb7d7ac966c60a7c1856
Executive archive SHA-256	b97bf1499fb82b6544c0a50f14e757ceef5e094d00c07f9f8c4633b87b219454

Use of this document. This package presents computational translational evidence from the PRISM-4D engine and downstream analysis layers. It is *not* a clinical-effect claim, *not* a patient-response prediction, and *not* experimental validation. Assay recommendations are falsification gates, not confirmation ceremonies. PROJECTED and HYPOTHESIZED items require wet-lab falsification before any biological assertion.

Epistemic legend. Every claim, table row, and figure inherits exactly one of the following classes, which governs how it may be cited:

- **OBSERVED** direct tensor measurement from PRISM-4D engine outputs.
- **DERIVED** deterministic transform of observed tensors.
- **INFERRED** multi-tensor interpretation; not a single-tensor measurement.
- **PROJECTED** translational extrapolation beyond simulated conditions.
- **HYPOTHESIZED** requires wet-lab falsification before any biological claim.

1 Section 1 — Master Data-Room Index

2 PRISM GLP-1R M2 Master Data Room

Campaign ID: glp1r_aleniglipron

Release date: 2026-05-23

Campaign Merkle Root Hash:

522dfcdd3179d4f0084938b97161bcb94ccbde0d2e29d111860ec7e45f8040c0

2.1 1. Executive Dossier

- Pharmacological Dynamics Intelligence Report: M2_Pharmacological_Dynamics_Intelligence_Report.md
- Enterprise Positioning Summary: ENTERPRISE_POSITIONING_SUMMARY.md
- Claim Falsification Graph: claim_falsification_graph.json

2.2 2. Interactive 3D Visualizer

Static visualizer entry point: campaigns/glp1r_aleniglipron/visualizer_app/index.html

The visualizer separates observed tensors from derived, inferred, projected, and hypothesized layers through the epistemic overlay controls.

2.3 3. Medicinal Chemistry Action Center

Top zero-shot replacement candidates are projected chemistry hypotheses generated by rigid-body grafting and tensor-field scoring.

Rank	Anchor	SMILES	SA Score	Pi Comple- ment	Pi Clash	Projected Durability Improvement
1	CAL- ANCHOR- 000453	<chem>Cc1cc(N2CCc3nn(-c4cc(C)c(F)c(C)c4)c(-c4cccc4)c3[C@H]2C)cc(C)c1F</chem>	3.35	14.6104	0.0000	73.27%
2	CAL- ANCHOR- 000405	<chem>Cc1cc(N2CCc3nn(-c4cc(C)c(F)c(C)c4)c(-c4cccc4)c3C2C)cc(C)c1F</chem>	3.35	14.3725	0.0000	72.69%
3	CAL- ANCHOR- 000338	<chem>Cc1cc(-c2c3c(nn2-c2cccc2)CCN(c2cc(C)c(F)c(C)c2)[C@@H]3C)cc(C)c1F</chem>	3.36	14.2591	0.0000	72.41%
4	CAL- ANCHOR- 000177	<chem>Cc1cc(-c2c3c(nn2-c2cc(C)c(F)c(C)c2)C CN(c2cccc2)[C@H]3C)cc(C)c1F</chem>	3.36	14.1984	0.0000	72.25%
5	CAL- ANCHOR- 000212	<chem>Cc1cc(-c2c3c(nn2-c2cccc2)CCN(c2cc(C)c(F)c(C)c2)C3C)cc(C)c1F</chem>	3.36	14.1778	0.0000	72.20%
6	CAL- ANCHOR- 000142	<chem>Cc1cc(-c2c3c(nn2-c2cc(C)c(F)c(C)c2)C CN(c2cccc2)[C@@H]3C)cc(C)c1F</chem>	3.36	14.1653	0.0000	72.17%
7	CAL- ANCHOR- 000107	<chem>Cc1cc(-c2c3c(nn2-c2cc(C)c(F)c(C)c2)C CN(c2cccc2)C3C)cc(C)c1F</chem>	3.36	14.1614	0.0000	72.16%

Rank	Anchor	SMILES	SA Score	Pi Comple- ment	Pi Clash	Projected Durability Improvement
8	CAL- ANCHOR- 000473	<chem>Cc1cc(N2CCc3nn(-c4cc(C)c(F)c(C)c4)c(-c4ccccc4)c3[C@@H]2C)cc(C)c1F</chem>	3.35	14.1216	0.0000	72.06%
9	CAL- ANCHOR- 000302	<chem>CCP(=O)(CC)c1ccc(-n2ccn(-c3ccccc3)c2=O)cc1-c1cc(C)c(F)c(C)c1</chem>	3.09	13.9999	0.0000	71.75%
10	CAL- ANCHOR- 000373	<chem>Cc1cc(-c2c3c(nn2-c2ccccc2)CCN(c2cc(C)c(F)c(C)c2)[C@@H]3C)cc(C)c1F</chem>	3.36	13.9889	0.0000	71.72%

Dynamic pharmacophore/SAR contingency rows:

2.3.1 Universal Growth Vectors

No universal growth vectors met the conserved thermally_activated cavity criterion. All current exit vectors exhibit high epistemic uncertainty (U_{pose}) and require scaffold rigidification before expansion.

2.3.2 Epistemic Mirage Vectors

Rule	Vector	Edge	Atom	U_pose	Threshold	Rationale
SAR-1222	Vector 1222	TYR152 -> TYR148	C22	0.0553	0.0251	Defer expansion at Vector 1222; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.

Rule	Vector	Edge	Atom	U_pose	Threshold	Rationale
SAR-1417	Vector 1417	TYR148 -> TYR152	C22	0.0553	0.0251	Defer expansion at Vector 1417; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.
SAR-1416	Vector 1416	TYR148 -> TYR152	C22	0.0553	0.0251	Defer expansion at Vector 1416; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.
SAR-1415	Vector 1415	TYR148 -> TYR152	C22	0.0553	0.0251	Defer expansion at Vector 1415; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.
SAR-1224	Vector 1224	TYR152 -> TYR148	C22	0.0553	0.0251	Defer expansion at Vector 1224; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.

Rule	Vector	Edge	Atom	U_pose	Threshold	Rationale
SAR-1223	Vector 1223	TYR152 -> TYR148	C22	0.0553	0.0251	Defer expansion at Vector 1223; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.
SAR-1422	Vector 1422	TYR148 -> TYR152	F24	0.0515	0.0251	Defer expansion at Vector 1422; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.
SAR-1423	Vector 1423	TYR148 -> TYR152	F24	0.0515	0.0251	Defer expansion at Vector 1423; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.
SAR-1421	Vector 1421	TYR148 -> TYR152	F24	0.0515	0.0251	Defer expansion at Vector 1421; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.

Rule	Vector	Edge	Atom	U_pose	Threshold	Rationale
SAR-1230	Vector 1230	TYR152 -> TYR148	F24	0.0515	0.0251	Defer expansion at Vector 1230; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.
SAR-1229	Vector 1229	TYR152 -> TYR148	F24	0.0515	0.0251	Defer expansion at Vector 1229; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.
SAR-1228	Vector 1228	TYR152 -> TYR148	F24	0.0515	0.0251	Defer expansion at Vector 1228; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.
SAR-1655	Vector 1655	PHE143 -> ILE147	C37	0.0488	0.0251	Defer expansion at Vector 1655; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.

Rule	Vector	Edge	Atom	U_pose	Threshold	Rationale
SAR-1654	Vector 1654	PHE143 -> ILE147	C37	0.0488	0.0251	Defer expansion at Vector 1654; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.
SAR-1653	Vector 1653	PHE143 -> ILE147	C37	0.0488	0.0251	Defer expansion at Vector 1653; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.
SAR-1678	Vector 1678	PHE143 -> ILE147	C45	0.0468	0.0251	Defer expansion at Vector 1678; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.
SAR-1677	Vector 1677	PHE143 -> ILE147	C45	0.0468	0.0251	Defer expansion at Vector 1677; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.

Rule	Vector	Edge	Atom	U_pose	Threshold	Rationale
SAR-1676	Vector 1676	PHE143 -> ILE147	C45	0.0468	0.0251	Defer expansion at Vector 1676; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.
SAR-1708	Vector 1708	PHE143 -> ILE147	C55	0.0452	0.0251	Defer expansion at Vector 1708; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.
SAR-1707	Vector 1707	PHE143 -> ILE147	C55	0.0452	0.0251	Defer expansion at Vector 1707; atom-level pose uncertainty exceeds the ray-casting threshold and requires scaffold rigidification before growth-vector optimization.

2.3.3 Prohibited Rigidification Zones

Rule	Vector	Edge	Classification	Score	Clash integral	Shear integral	Rationale
SAR-1714	Vector 1714	PHE143 -> ILE147	prohibited_rigidification_zone	-8.4458	2.5098	5.9360	Halt rigidifying expansion at Vector 1714; the ray intersects a positive clash field near a receptor lock interface.

Rule	Vector	Edge	Classification	Score	Clash integral	Shear integral	Rationale
SAR-1328	Vector 1328	TYR152 -> TYR148	prohibited_rigidification_zone	-8.4458	2.5098	5.9360	Halt rigidifying expansion at Vector 1328; the ray intersects a positive clash field near a receptor lock interface.
SAR-1727	Vector 1727	PHE143 -> ILE147	prohibited_rigidification_zone	-8.2538	2.3853	5.8685	Halt rigidifying expansion at Vector 1727; the ray intersects a positive clash field near a receptor lock interface.
SAR-1189	Vector 1189	TYR152 -> TYR148	prohibited_rigidification_zone	-8.2254	2.6790	5.5464	Halt rigidifying expansion at Vector 1189; the ray intersects a positive clash field near a receptor lock interface.
SAR-1575	Vector 1575	PHE143 -> ILE147	prohibited_rigidification_zone	-8.2254	2.6790	5.5464	Halt rigidifying expansion at Vector 1575; the ray intersects a positive clash field near a receptor lock interface.
SAR-1623	Vector 1623	PHE143 -> ILE147	prohibited_rigidification_zone	-8.1544	2.3719	5.7825	Halt rigidifying expansion at Vector 1623; the ray intersects a positive clash field near a receptor lock interface.

Rule	Vector	Edge	Classification	Score	Clash integral	Shear integral	Rationale
SAR-1191	Vector 1191	TYR152 -> TYR148	prohibited_rigidification_zone	-7.9600	2.3868	5.5732	Halt rigidifying expansion at Vector 1191; the ray intersects a positive clash field near a receptor lock interface.
SAR-1577	Vector 1577	PHE143 -> ILE147	prohibited_rigidification_zone	-7.9600	2.3868	5.5732	Halt rigidifying expansion at Vector 1577; the ray intersects a positive clash field near a receptor lock interface.
SAR-1200	Vector 1200	TYR152 -> TYR148	prohibited_rigidification_zone	-7.9466	2.9252	5.0214	Halt rigidifying expansion at Vector 1200; the ray intersects a positive clash field near a receptor lock interface.
SAR-1586	Vector 1586	PHE143 -> ILE147	prohibited_rigidification_zone	-7.9466	2.9252	5.0214	Halt rigidifying expansion at Vector 1586; the ray intersects a positive clash field near a receptor lock interface.
SAR-1190	Vector 1190	TYR152 -> TYR148	prohibited_rigidification_zone	-7.9340	2.6767	5.2574	Halt rigidifying expansion at Vector 1190; the ray intersects a positive clash field near a receptor lock interface.

Rule	Vector	Edge	Classification	Score	Clash integral	Shear integral	Rationale
SAR-1576	Vector 1576	PHE143 -> ILE147	prohibited_rigidification_zone	-7.9340	2.6767	5.2574	Halt rigidifying expansion at Vector 1576; the ray intersects a positive clash field near a receptor lock interface.
SAR-1600	Vector 1600	PHE143 -> ILE147	prohibited_rigidification_zone	-7.9019	2.5233	5.3786	Halt rigidifying expansion at Vector 1600; the ray intersects a positive clash field near a receptor lock interface.
SAR-1646	Vector 1646	PHE143 -> ILE147	prohibited_rigidification_zone	-7.8535	2.4255	5.4280	Halt rigidifying expansion at Vector 1646; the ray intersects a positive clash field near a receptor lock interface.
SAR-1260	Vector 1260	TYR152 -> TYR148	prohibited_rigidification_zone	-7.8535	2.4255	5.4280	Halt rigidifying expansion at Vector 1260; the ray intersects a positive clash field near a receptor lock interface.
SAR-1263	Vector 1263	TYR152 -> TYR148	prohibited_rigidification_zone	-7.7664	2.3741	5.3923	Halt rigidifying expansion at Vector 1263; the ray intersects a positive clash field near a receptor lock interface.

Rule	Vector	Edge	Classification	Score	Clash integral	Shear integral	Rationale
SAR-1579	Vector 1579	PHE143 -> ILE147	prohibited_rigidification_zone	-7.7480	2.4172	5.3308	Halt rigidifying expansion at Vector 1579; the ray intersects a positive clash field near a receptor lock interface.
SAR-1193	Vector 1193	TYR152 -> TYR148	prohibited_rigidification_zone	-7.7480	2.4172	5.3308	Halt rigidifying expansion at Vector 1193; the ray intersects a positive clash field near a receptor lock interface.
SAR-1597	Vector 1597	PHE143 -> ILE147	prohibited_rigidification_zone	-7.6872	2.3698	5.3173	Halt rigidifying expansion at Vector 1597; the ray intersects a positive clash field near a receptor lock interface.
SAR-1572	Vector 1572	PHE143 -> ILE147	prohibited_rigidification_zone	-7.6299	2.4830	5.1469	Halt rigidifying expansion at Vector 1572; the ray intersects a positive clash field near a receptor lock interface.

2.4 4. Wet-Lab CRO Handoff

Each row is framed as a falsification gate rather than a confirmation request.

Assay	Condition	Residue	Peptide Window	Construct	Measurement Readout	Execution Instruction	Falsification Condition
HDX-MS	glp1r_5VEX_WT	ASN182:46	[41, 51]	WT-normalized apo and holo GLP-1R peptide-mapping panel.	Directional uptake asymmetry between Apo and Holo states.	Run HDX-MS as a falsification gate for the predicted shear-associated exposure asymmetry; target peptide window [41, 51].	Falsification Condition: Lack of WT-normalized uptake asymmetry in the targeted peptide window.
HDX-MS	glp1r_6X1A_WT	ASN182:148	[143, 153]	WT-normalized apo and holo GLP-1R peptide-mapping panel.	Directional uptake asymmetry between Apo and Holo states.	Run HDX-MS as a falsification gate for the predicted shear-associated exposure asymmetry; target peptide window [143, 153].	Falsification Condition: Lack of WT-normalized uptake asymmetry in the targeted peptide window.
HDX-MS	glp1r_6XOX_A316T	ASN182:144	[139, 149]	WT-normalized apo and holo GLP-1R peptide-mapping panel.	Directional uptake asymmetry between Apo and Holo states.	Run HDX-MS as a falsification gate for the predicted shear-associated exposure asymmetry; target peptide window [139, 149].	Falsification Condition: Lack of WT-normalized uptake asymmetry in the targeted peptide window.
HDX-MS	glp1r_6X1A_WT	PHE367:333	[328, 338]	WT-normalized apo and holo GLP-1R peptide-mapping panel.	Directional uptake asymmetry between Apo and Holo states.	Run HDX-MS as a falsification gate for the predicted shear-associated exposure asymmetry; target peptide window [328, 338].	Falsification Condition: Lack of WT-normalized uptake asymmetry in the targeted peptide window.
HDX-MS	glp1r_6XOX_WT	ASN182:144	[139, 149]	WT-normalized apo and holo GLP-1R peptide-mapping panel.	Directional uptake asymmetry between Apo and Holo states.	Run HDX-MS as a falsification gate for the predicted shear-associated exposure asymmetry; target peptide window [139, 149].	Falsification Condition: Lack of WT-normalized uptake asymmetry in the targeted peptide window.